LP is Exact - Summary so far

Min length r-Arborescence LP: $G = (V, E)$ directed

$\ell \in \mathbb{Z}^{E}_{\geq 0}$ - lengths

C columns correspond to arcs/edges // exp'ly many rows.

rows correspond to r-cuts

$$\begin{aligned} \min:\ & \ell^T x \\ \text{st:}\ & Cx \geq \mathbb{1} \\ & x \geq 0 \end{aligned} \qquad \begin{aligned} \max:\ & \mathbb{1}^T y \\ \text{st:}\ & y^T C \leq \ell^T \\ & y \geq 0 \end{aligned}$$

y a sol'n of max that also maximizes $\sum y_u \cdot |U|^2$

Ex: $\mathcal{F} = \{U \subseteq V \setminus \{r\} : y_u > 0\}$ Laminar

C' submatrix of C, corresponding to $U \in \mathcal{F}$.

Saw:

$$\begin{aligned} \max:\ & \mathbb{1}^T y \\ \text{st:}\ & y^T C \leq \ell^T \\ & y \geq 0 \end{aligned} \quad = \quad \begin{aligned} \max:\ & \mathbb{1}^T y \\ \text{st:}\ & y^T C' \leq \ell^T \\ & y \geq 0 \end{aligned}$$

C’ is TU

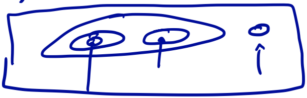
Lemma: C' is totally unimodular

pf: Choose a subset of rows of C'

$\mathcal{G} \subseteq \mathcal{F}$

For each $U \in \mathcal{G}$, define the height of U

by: $h(U) = \#$ of sets $T \in \mathcal{G}$ with $T \supseteq U$

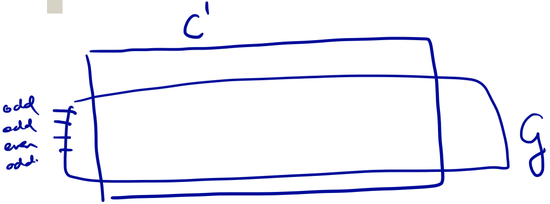


$$\mathcal{G}_{even} = \{U \in \mathcal{G} : h(U) \text{ is even}\}$$

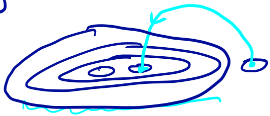
$$\mathcal{G}_{odd} = \{U \in \mathcal{G} : h(U) \text{ is odd}\}$$

$$\mathcal{G} = \mathcal{G}_{even} \cup \mathcal{G}_{odd}$$

C' is TU — continued



Claim: For any edge e of $G = (V, E)$ the number of sets in $\mathcal{G}_{odd}$ entered by e differs from the number of sets in $\mathcal{G}_{even}$ entered by e, by at most 1.

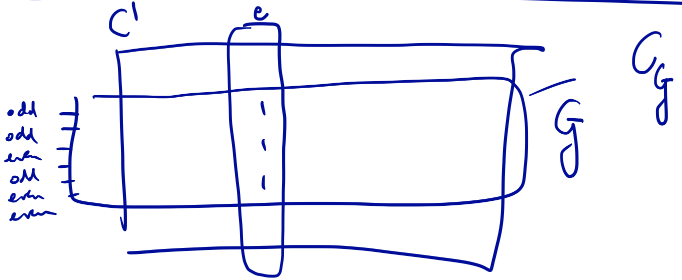


C' is TU — continued

pf of claim: rows correspond to r-cuts and critically, $\mathcal{G}$ is laminar.

Exercise: Fill in the details of the proof of the claim.

$\Rightarrow C'$ is TU.



Conclusion

C' is TU $\Rightarrow$
$$\begin{aligned} \max:\ & y^T \mathbb{1} \\ \text{st:}\ & y^T C' \leq \ell^T \\ & y \geq 0 \end{aligned}$$

has an integral opt solution

$$\Rightarrow \quad \begin{aligned} \max:\ & y^T \mathbb{1} \\ \text{st:}\ & y^T C \leq \ell^T \\ & y \geq 0 \end{aligned}$$

has integral opt. solution

by Hoffman & Kruskal: $\Rightarrow$ TDI

$\Rightarrow$ Primal has integral solution $\forall \ell$ integral.

Why is $\mathcal{F}$ Laminar?

C ~~is not~~ need not be totally unimodular